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# Nonlinear chiroptical response in lithium niobate metasurface driven by quasi-bound states in the continuum

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## ABSTRACT

Quasi-bound states in the continuum (QBIC) can significantly localize the light field and enhance light–matter interactions at the nanoscale, providing a platform for high-Q chiral light response and promoting nonlinear effects of materials. In this work, we numerically study the chiral linear and nonlinear light responses of the QBIC lithium niobate (LN) metasurface and achieve chirality modulation. The designed metasurface consists of LN nanobar dimers, and the chiral QBIC mode is excited by breaking the in-plane and out-of-plane symmetries of the structure, with the circular dichroism (CD) value and Q-factor reaching 0.92 and $1.24 \times 10^4$, respectively. Then, we investigate the second harmonic generation (SHG) of this device. The conversion efficiency of SHG under right circularly polarized pumping reaches $7.3 \times 10^{-3}$, which is more than three orders of magnitude higher than that under the left circularly polarized pumping. The corresponding CD value of SHG reaches 0.99. In addition, by introducing phase change materials, we study the active modulation of the chiroptical response. Our results provide a crucial route for high-quality chiral light sources.

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Chirality is a fundamental property of an object, referring to the inability of an object to be made identical to its mirror image through translation or rotation.[1–4] Interaction of chiral objects with circularly polarized light induces chiral optical responses,[5] such as optical rotation and circular dichroism (CD).[6,7] It is noted that the chiroptical response of natural materials is usually very weak.[8] Recently, bound states in the continuum (BICs), a nonradiative mode that can greatly enhance light–matter interaction,[9–15] have been used to enhance chiroptical effects. By breaking the in-plane and out-of-plane symmetries of the structure, chiral quasi-bound states in the continuum (QBICs) of the resonant metasurface are excited, providing a platform for achieving high-quality factor (Q-factor) resonances and large chiral responses.[16–18] For example, in 2020, Gorkunov *et al.* demonstrated that a rotationally symmetric dielectric bar chiral metasurface can achieve ultrahigh CD under QBIC mode.[1] In the following year, Overvig *et al.* introduced chiral perturbations, enabling the QBIC to encode arbitrary elliptical polarization states and geometric phase engineering.[19] In 2023, Chen *et al.* proposed a tilted trapezoidal nanohole metasurface with the first experimental observation of intrinsic chiral QBIC, with the Q-factor and the CD values reaching 2663 and 0.93,[20] respectively.

In recent years, chiral QBICs, because of their high Q-factor, are widely used in applications such as chiral sensing[21] and chiral light emission.[22] In addition, chiral QBIC metasurfaces are also used in nonlinear optics.[23,24] The nonlinear CD is defined as the strong difference in harmonic conversion efficiency between right circularly polarized (RCP) and left circularly polarized (LCP) light.[25–27] For example, in 2022, Shi *et al.* investigated a planar chiral QBIC metasurface formed by an α-Si double-sided scythe structure, which theoretically achieved an efficient third harmonic CD value of 0.99.[25] In 2023, Koshelev *et al.* designed a chiral metasurface with L-shaped asymmetric Si. QBIC modes can generate large third harmonic signals, and the experimentally measured third harmonic CD poles are −0.771 and 0.918.[26] In 2024, Cai *et al.* proposed a Si elliptic semicircular disk dimer metasurface that excites chiral QBIC by breaking the in-plane and out-of-plane symmetries of the structure and achieved large third-order and fifth-order nonlinear harmonic CD.[27] Except for the silicon material mentioned earlier, in recent years, another important nonlinear material, lithium niobate (LN), has been widely considered in nonlinear optics.[28,29] LN belongs to the 3 m point group, it has a large second-order nonlinear coefficient and a wide optical transparency


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window (350 nm–5 $\mu$m),[30,31] which provides a platform for efficient second harmonic generation (SHG).[32–37] It is noted that these SHG studies are mainly focused on the nonchiral QBIC nanostructures[32–36] and weak guide mode resonance nanostructure.[37] At present, the research on chiral light sources based on LN QBIC metasurface has not yet been reported.

In this work, we numerically investigate chiral linear and nonlinear responses of a LN metasurface that supports the QBIC mode. The designed metasurface is composed of periodic an array of LN nanobar dimers. By breaking the in-plane and out-of-plane symmetries of the structure, the chiral QBIC is excited, which is sensitive to the RCP light, while the response is weaker under the LCP incidence, the numerically calculated linear CD value and Q-factor are 0.92 and $1.24 \times 10^4$, respectively. Moreover, the SHG conversion efficiency under the RCP pump is more than three orders of magnitude that of the LCP pump, and corresponding CD value of SHG reaches 0.99. We also investigate active modulation of chiroptical response by introducing a phase change material. Our results provide a route for high-quality chiral and nonlinear light sources.

The designed chiral nonlinear LN metasurface is shown in Fig. 1(a). The top layer of the device is LN nanobar dimers, with a refractive index of $n_x = n_z = n_o = 2.2264$, $n_y = n_e = 2.1506$.[38] The middle layer is Si with a thickness of 220 nm, and the bottom layer is $SiO_2$. Figure 1(b) shows the geometric transformation of the unit cell, which illustrate the achievement of chiral QBIC. The period is set to $P = 1000$ nm, the thickness of the LN nanobars $h = 200$ nm, the width $a = 250$ nm, the center spacing of the two nanobars $d = 512$ nm, and the length $b = 484$ nm. First, the unit cell with a pair of parallel and same rectangular nanobars is marked by model I, which support the BIC mode. With the introduction of the deflection angle $\theta$, the nonchiral QBIC mode is excited, as shown in operation II in Fig. 1(b). To create the maximum chiral QBIC, we maintain the deflection angle $\theta$ of the two nanobars at 23° and turn one of the nanobars sideways, as shown for operation III in Fig. 1(b). These nanobars of different heights is possibly achieved through techniques such as nesting,[39] physical vapor deposition,[40] or electron beam lithography.[41,42] Then, we calculate the band structure, Q-factors in the momentum space, polarization state in the far field of the pattern I, as shown in Figs. 1(c)–1(e). The BIC has an infinite Q-factor at the Γ point, and the Q-factors decrease significantly with the in-plane wave vector component, the polarization state in the far field has a topological charge of −1. These results demonstrate that model I supports the BIC mode. Figure 1(f) shows the Q-factors and the resonance wavelength as a function of the deflection angle $\theta$ of the metasurface, where the blue line denotes the resonance wavelength with different $\theta$, the red dot-line represents the calculated Q-factors. It is clearly seen that the Q-factors decrease with the increase in the $|\theta|$, which demonstrates that the QBIC mode is excited and corresponds to operation II. The all optical properties of the metasurface are evaluated by the finite element method with commercial software COMSOL Multiphysics.

Next, the chiral response of the metasurface is analyzed, which is characterized by applying the Jones matrix to the transmission coefficient ($t_{ij}$) under circular polarization conditions,[25,43]

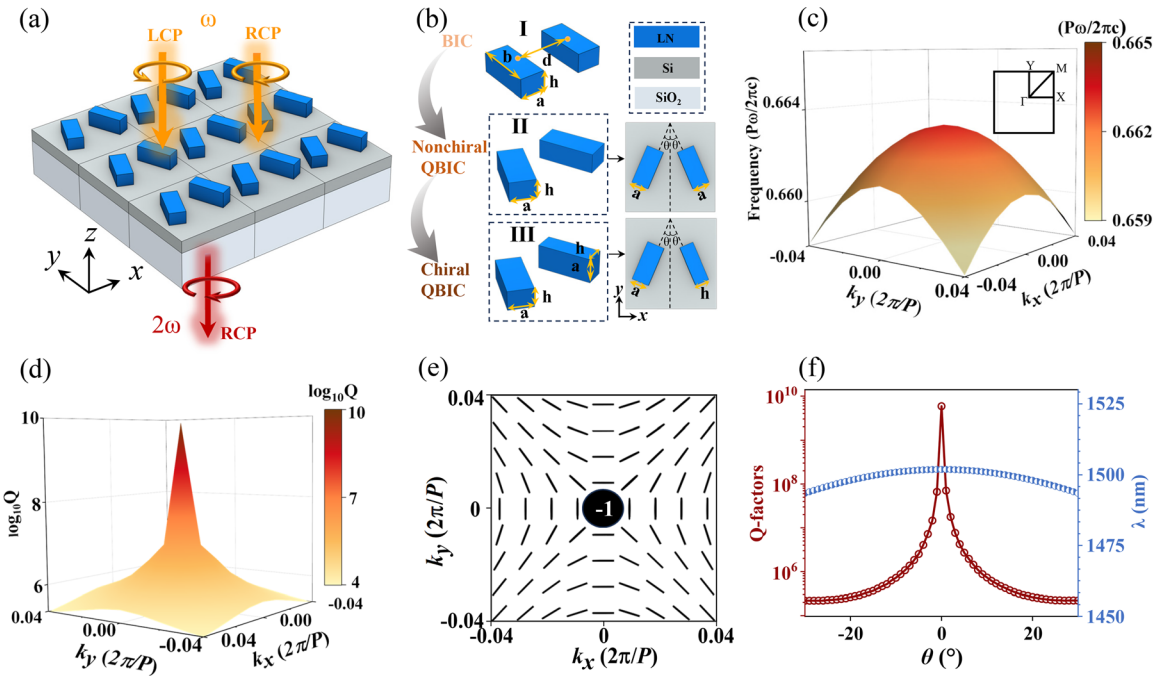



**FIG. 1.** (a) Schematic diagram of SHG of chiral nonlinear metasurface structure. Circularly polarized light with frequency $\omega$ is incident along the $z$ direction, perpendicular to the surface. (b) Left: schematic of the symmetry-breaking processes that transfer the BIC to chiral Q-BIC. Right: the insets show materials of structure (above), top view of the unit cell, which introduced deflection angle $\theta$ (middle) and turn one of the nanobars sideways (below). The deflection angle is defined by the $\theta$ between the x axis and the long axis of the nanobar. (c) Band structure for nanobar arrays. (d) Radiating Q-factors plotted in the momentum space, and the Q-factor tend to infinity at Γ point ($k_x = k_y = 0$). (e) Polarization state in the far field has a topological charge of −1. (f) Q-factors and wavelength for different deflection angle $\theta$.

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$$J_{cicr} = \begin{pmatrix} t_{RR} & t_{RL} \\ t_{LR} & t_{LL} \end{pmatrix}, \quad (1)$$

where the subscripts R and L represent RCP and LCP, respectively. $t_{ij}$ ($i = L, R$; $j = L, R$) denotes the polarized transmission coefficient of the output $i$ relative to the input $j$. The transmittance is given by the equation $T_{ij} = |t_{ij}|^2$. During the application process, the circularly polarized light with Jones vectors $\begin{pmatrix} 1 \\ i \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -i \end{pmatrix}$ are denoted as RCP and LCP light, respectively. The CD takes up the majority of the chiroptical phenomenon and provides a judgment of the chiral intensity, which is defined as the transmittance difference between different Jones matrix components under RCP and LCP incidence in circularly polarized bases. The calculation formula is expressed as[44]

$$CD = \frac{(T_{RR} + T_{LR}) - (T_{LL} + T_{RL})}{(T_{RR} + T_{LR}) + (T_{LL} + T_{RL})}. \quad (2)$$

The $T_{RR}$, $T_{LR}$, $T_{LL}$, $T_{RL}$, and CD value at different circularly polarized light are calculated, as shown in Fig. 2(a). As for the chiral QBIC mode at 1492.8 nm, it is evident that the contribution of cross-polarized transmission is almost zero, and a distinct transmission resonance is observed only for the RCP incidence, while remaining weakly coupled with the LCP incidence. Therefore, a large CD value of 0.92 is achieved, and the Q-factor is $1.24 \times 10^4$. Subsequently, the far field scattering power of multiple dipoles is calculated using the multipole decomposition method.[45–47] The multipole decomposition results are shown in Figs. 2(b) and 2(c), which are under the RCP and LCP irradiation, respectively. It can be clearly seen that the far-field scattering power of RCP incident light is two orders of magnitude higher than that of LCP light. Furthermore, the magnetic dipole oscillation has always been dominant, which indicates that the resonance modes excited by the two polarized lights are the same. The insets show the electric field distribution characteristics at the resonance wavelength of RCP and LCP, the results show that the field intensity excited by RCP

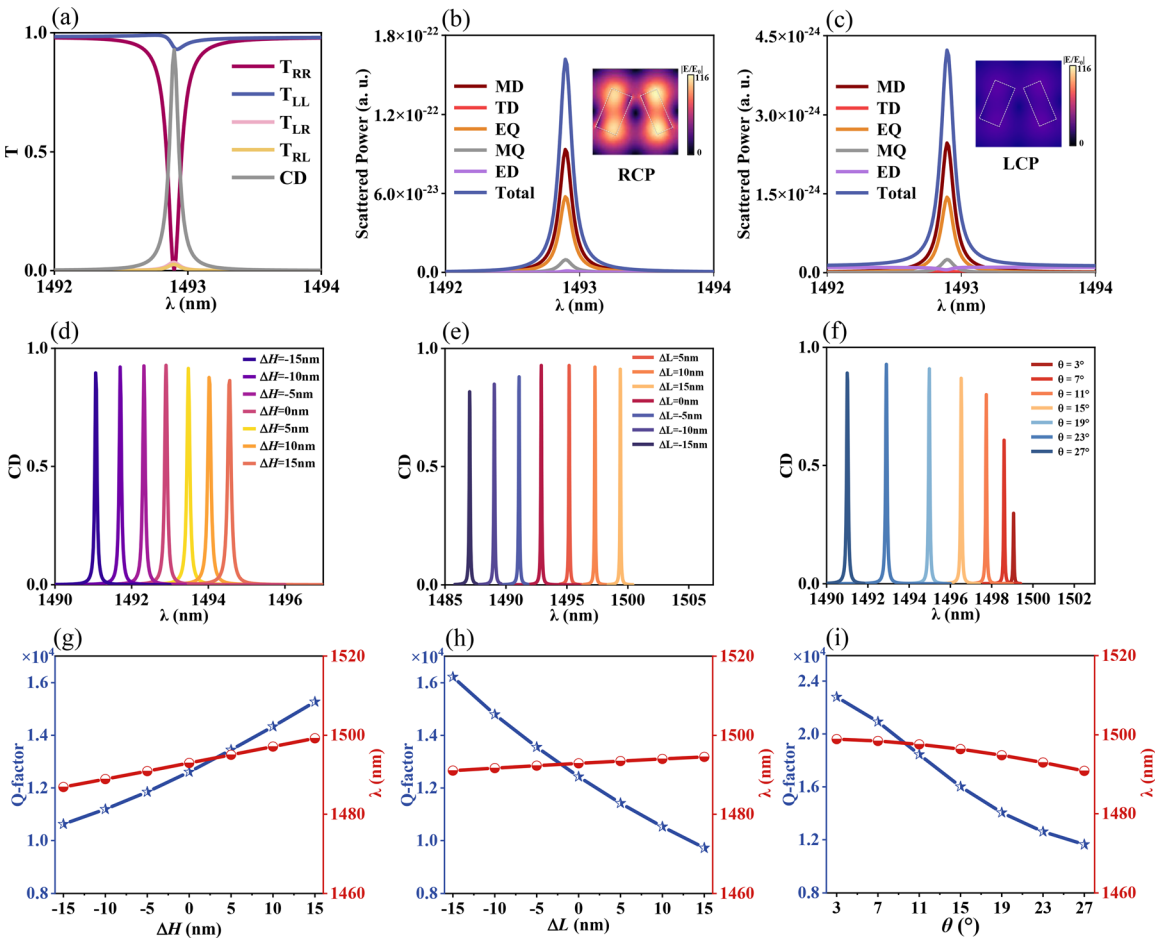



**FIG. 2.** (a) $T_{RR}$, $T_{LR}$, $T_{LL}$, and $T_{RL}$ are the transmission components defined by the Jones matrix and the calculated CD spectrum. (b) and (c) Far-field scattering power of the multipole of LCP light and RCP light excitation, respectively. The insets show electric field intensity distribution in the $x$-$y$ plane under RCP and LCP excitation, with the color shades representing the magnitude of the field intensity, where yellow represents the maximum, and purple, the minimum. (d)–(f) The CD spectra at different thicknesses $\Delta H$, lengths and widths $\Delta L$, and angles $\theta$. (g)–(i) The Q-factors and wavelengths for different thicknesses $\Delta H$, lengths and widths $\Delta L$, and angles $\theta$.

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is significantly stronger than that of LCP. Moreover, the CD values of the two nanobars at different thicknesses $\Delta H$, lengths and widths $\Delta L$, and angles $\theta$ are calculated. As shown in Figs. 2(d) and 2(g), when the thickness of the two nanobars vary from −15 to 15 nm, the linear CD values consistently remain above 0.86, while the Q-factor and wavelength are both gradually increasing. When the length and width of the two nanobars simulaneously vary from −15 to 15 nm, as shown in Figs. 2(e) and 2(h), the linear CD values first increase and then decrease, with all values staying above 0.82. During this process, the Q-factor gradually decreases, while the wavelength remains relatively robust. By changing the deflection angle of the two nanobars, as shown in Fig. 2(f), the CD values are also gradually increasing when the angle increases from 3° to 23°, after which the CD value decreases. Additionally, as shown in Fig. 2(i), the Q-factor and wavelength are gradually decreasing. Therefore, by changing the thicknesses $\Delta H$ or lengths and widths $\Delta L$ of LN nanobars, the CD values are more stable; when the deflection angles $\theta$ decrease, the CD values decrease.

In nonlinear optics, the chiral QBIC metasurface opens possibilities for harmonic generation by enhancing the material's nonlinear response. Following that, the nonlinear conversion efficiency is calculated to evaluate the nonlinear effect of the designed chiral metasurface. For SHG, the conversion of fundamental frequency pump light to second harmonic power radiation can be expressed as follows: $P(2\omega) \propto [P(\omega)\kappa_1(\omega)Q_1L_1(\omega)]^2\kappa_{12}Q_2L_2(2\omega)\kappa_2(2\omega)$,[48,49] where $P(\omega)$ is the incident power of the pump light. $Q_1$ and $Q_2$ are the Q factors of the resonances at the fundamental and the second harmonic frequencies. $L_1(\omega)$ and $L_2(2\omega)$ are spectral overlap and mismatch factors. $\kappa_1(\omega)$ and $\kappa_2(\omega)$ are coupling and outcoupling factors at the pump and second harmonic frequencies. $\kappa_{12}$ is the cross coupling coefficient that depends on the second-order nonlinear susceptibility tensor of the nonlinear material. Additionally, the relationship between the second harmonic polarization intensity formed inside LN and the pump photoelectric field satisfy the following equation:[38,49]

$$\begin{bmatrix} P_x^{SH} \\ P_y^{SH} \\ P_z^{SH} \end{bmatrix} = 2\varepsilon_0 d_{il} \begin{bmatrix} E_x^2 \\ E_y^2 \\ E_z^2 \\ 2E_yE_z \\ 2E_xE_z \\ 2E_xE_y \end{bmatrix} \tag{3}$$

and

$$d_{il} = \begin{bmatrix} 0 & 0 & 0 & 0 & d_{22} & d_{31} \\ d_{31} & d_{31} & d_{33} & 0 & 0 & 0 \\ d_{22} & 0 & -d_{22} & d_{31} & 0 & 0 \end{bmatrix}, \tag{4}$$

where $d_{33} = -19.5$ pm/V, $d_{31} = -3.5$ pm/V, and $d_{22} = 1.9$ pm/V.[29]

In order to gain a deeper understanding of the relationship between the harmonic power and the incident optical power, as shown in Figs. 3(a) and 3(b), the SHG powers for different pump intensities under the RCP and LCP incident light are calculated, respectively. It can be clearly seen that the scattering power of SHG gradually increases as the pump light power increases, and the SHG scattering power under RCP light being three orders of magnitude higher than that under LCP light. To further confirm the nonlinear optical

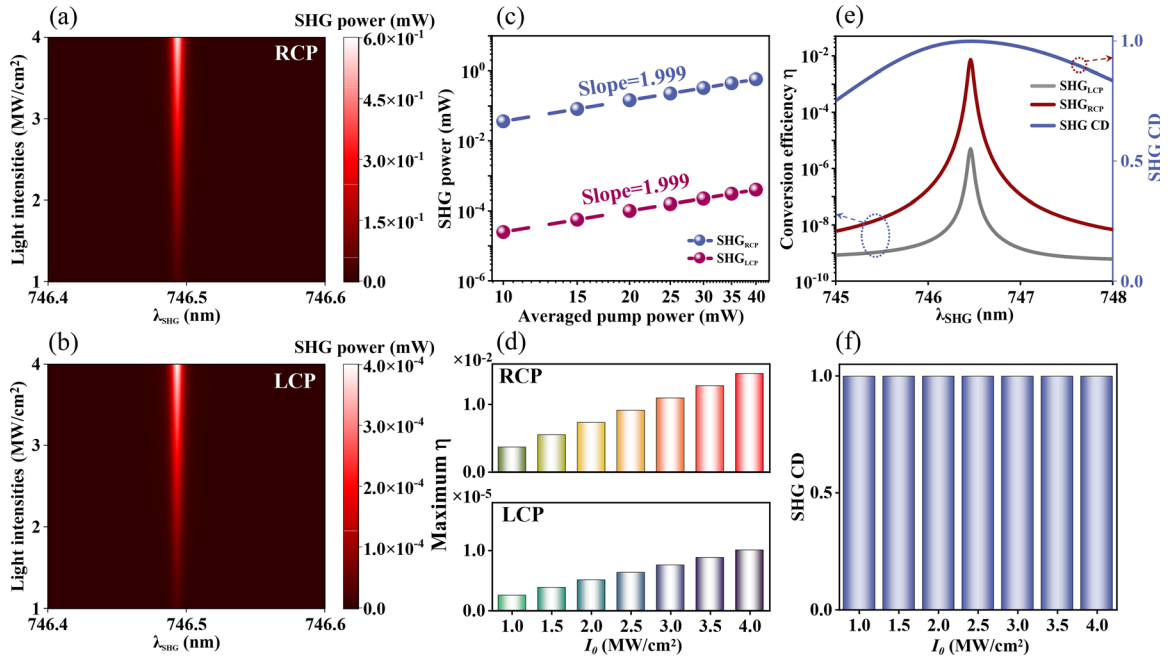



**FIG. 3.** (a) and (b) SHG powers mapping for different pump light intensities under RCP and LCP incident light. (c) The relationship between the SHG powers and the average pump powers under RCP and LCP incident light, the slope values of the fitting straight lines are 1.999. (d) Maximum conversion efficiency of SHG for different pump light intensities under RCP and LCP incident light. (e) Simulated SHG conversion efficiencies under different circular polarized light incidence and their corresponding SHG CD when incident light intensity is 2 MW/cm$^2$. (f) CD values of SHG at different incident light intensities.

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properties of the SHG, we plot the relationship between the average pump power and the average SHG power as in Fig. 3(c). Taking the logarithm of the horizontal and vertical coordinates, the slopes of the fitted straight lines are 1.999, which are close to the theoretical values of 2 and fully confirm that the metasurface generate the SH signal. The SHG conversion efficiency is expressed as $\eta_{SHG} = P_{peak}^{SHG}/P_{peak}^{FW}$,[50] where $P_{peak}^{SHG}$ is the peak scattered power at the second harmonic, and $P_{peak}^{FW}$ is the incident peak power at the fundamental frequency. The maximum conversion efficiencies exhibit a synchronized increase with the intensities of light, as shown in Fig. 3(d). Furthermore, the SHG conversion efficiencies of the LN metasurface under RCP light is much higher than those under LCP light when pump light intensities range from 1 to 4 MW/cm$^2$. Then, in order to analyze the difference in conversion efficiency between RCP and LCP for SHG, we calculate the CD of SHG, which is defined as the rate of the difference in SHG conversion efficiency under RCP and LCP irradiation, and the CD expressions for SHG is $CD_{SHG} = \frac{\eta_{RCP}^{2\omega} - \eta_{LCP}^{2\omega}}{\eta_{RCP}^{2\omega} + \eta_{LCP}^{2\omega}}$.[23] Figure 3(e) shows the line shape of the SHG and corresponding SHG CD under RCP and LCP light when the pump light intensity is set to 2 MW/cm$^2$. The $\eta_{SHG}$ reaches $7.3 \times 10^{-3}$ and $5 \times 10^{-6}$ under the RCP and LCP light, respectively, and their corresponding CD value of the SHG reaches 0.99. Figure 3(f) shows the CD values of SHG when the incident light intensity increases from 1 to 4 MW/cm$^2$. It can be seen that the SHG CD is stable at around 0.99.

Furthermore, the active chiral response of the LN metasurface is investigated by introducing the phase change material $Sb_2Se_3$ to replace Si, as shown in Fig. 4(a), which can achieve modulation by altering the QBIC mode field distribution. The phase change of phase change materials can usually be achieved by temperature[51] or lasers,[52,53] and so on. The permittivities of a-$Sb_2Se_3$ and c-$Sb_2Se_3$ at 400–1800 nm wavelength can be obtained from Refs. 52 and 54. Figure 4(b) shows the real and imaginary parts of the refractive index for $Sb_2Se_3$ at degrees of crystallinity corresponding to crystallization degree $m = 0$, 0.5, and 1, respectively. It is clear that the loss of the $Sb_2Se_3$ is almost 0 in the wavelength range of 1200–1800 nm. The characteristics of $T_{RCP}$, $T_{LCP}$, and CD values in different crystallinity is shown in Fig. 4(c). When the crystallization degrees are 0, 0.5, and 1, the CD values are 0.11, 0.86, and 0.67, respectively. It can be seen that relative to the other two degrees of crystallinity, the CD value is greatest when the degree of crystallinity is 0.5. This is because the refractive index of the phase change material at this time is close to the refractive index of the silicon that was previously replaced. Meanwhile, SHG and $CD_{SHG}$ are calculated for the corresponding crystallinity, when the

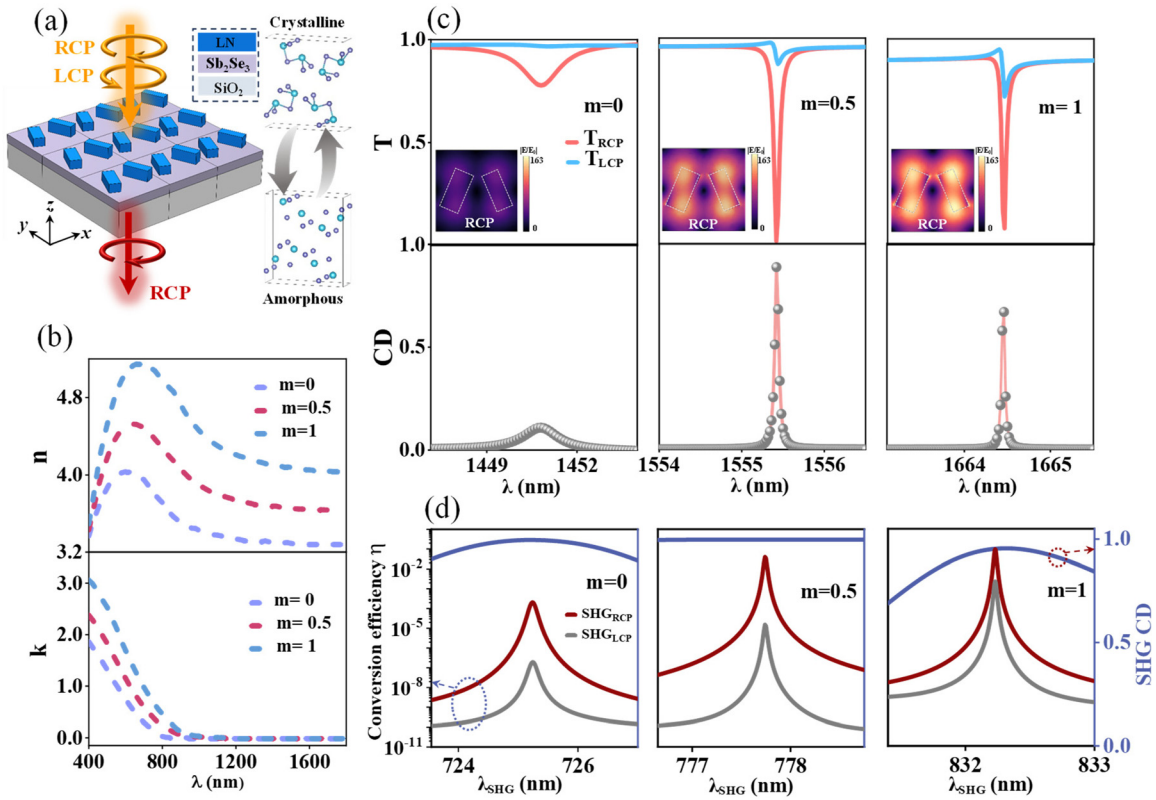



**FIG. 4.** Schematic diagram of chiral nonlinear metasurface structure with $Sb_2Se_3$ thin film. (b) The refractive index and absorption loss of $Sb_2Se_3$ when the wavelength is varied from 400 to 1800 nm with crystallization degree $m = 0$, 0.5, and 1. (c) Spectral characteristics of $T_{RCP}$, $T_{LCP}$, and CD values when degrees of crystallinity are $m = 0$, 0.5, and 1, respectively. The insets show electric field intensity distribution in the $x$–$y$ plane under RCP excitation, with the color shades representing the magnitude of the field intensity, where yellow represents the maximum, and purple, the minimum. (d) SHG conversion efficiencies of RCP and LCP excitation when incident light intensity is 2 MW/cm$^2$, and their corresponding SHG CD is under varying degrees of crystallinity.

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incident light intensity is 2 MW/cm$^2$, as shown in Fig. 4(d). We can observe that the SHG conversion efficiency increases gradually as the crystallinity increases. The refractive index of the phase change material increases during the increase in the crystallinity, which enhances the localized light field within the device and thereby enhances the conversion efficiency of SHG. The SHG CD values corresponding to different degrees of crystallinity are 0.99, 0.98, and 0.95, respectively. This is because the difference between the SHG conversion efficiencies at RCP and LCP light incidence is gradually decreasing, which leads to a gradual decrease in SHG CD as well. Therefore, the modulation of chiroptical response can be achieved by using phase change material.

In summary, we investigate the chiral linear and nonlinear properties of LN metasurfaces with a QBIC mode and achieve modulation by introducing a phase change material. The designed metasurface consists of LN nanobar dimers, and the chiral QBIC mode is excited by breaking the in-plane and out-of-plane symmetries of the structure. We achieve a Q-factor of $1.24 \times 10^4$ and a CD value of 0.92 when one of the nanobars is turned sideways, and the deflection angle $\theta$ of the two nanobars is set at 23°. On this basis, we further investigate the SHG conversion efficiency of the chiral metasurface. The SHG conversion efficiency reaches up to $7.3 \times 10^{-3}$ and $5 \times 10^{-6}$ under the RCP and LCP light, respectively, and the CD value of the SHG reaches 0.99. Additionally, the active modulation of chiroptical response can be achieved by adjusting the crystallinity of the $Sb_2Se_3$. Our results provide tunable optical chirality and introduce ideas for exciting high-quality chiral light and advancing nonlinear chiral optics.

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## AUTHOR DECLARATIONS

### Conflict of Interest

The authors have no conflicts to disclose.

### Author Contributions

**Hong Duan:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Haoxuan He:** Data curation (equal); Investigation (equal); Methodology (equal); Supervision (equal); Validation (equal); Writing – original draft (equal). **Yingfei Yi:** Conceptualization (equal); Formal analysis (equal); Methodology (equal); Supervision (equal). **Lulu Wang:** Investigation (equal); Supervision (equal); Writing – review & editing (equal). **Ying Zhang:** Formal analysis (equal); Supervision (equal); Writing – review & editing (equal). **Xia Yan:** Investigation (equal); Supervision (equal). **Jing Huang:** Funding acquisition (equal); Investigation (equal); Supervision (equal). **Chaobiao Zhou:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Project administration (equal); Resources (equal); Software (equal); Supervision (equal); Visualization (equal); Writing – review & editing (equal).

## DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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